

DSCI-28 — Interpretable Cross-Country Poverty Estimation from Survey Data

Canonical ID DSCI-28	Dominant specialization Data Science & Big Data Analytics
Technical domain Data Science and Big Data Analytics	Strongest impact Industry Impact
Reference source: DrivenData · DrivenData / World Bank https://www.drivendata.org/competitions/50/worldbank-poverty-prediction/	

READ THIS BEFORE CHOOSING THE PROJECT

Two-semester capstone project charter

CP1 + CP2 · complete engineering system

PROBLEM TO SOLVE

Poverty-targeting programmes decide who receives assistance using models built on national household surveys, so these models must generalise across countries with very different wealth distributions and. Critically — must be interpretable enough to defend in policy settings where people are excluded from benefits by their output. The consequence is incomplete, incorrect or non-reproducible evidence reaching downstream decisions in Media, Entertainment & Social Platforms.

WHO BENEFITS AND WHAT OWNERSHIP IS ESTABLISHED

Primary beneficiaries are World Bank users, maintainers and technical reviewers; the cited record identifies World Bank as the problem issuer or upstream custodian, not as the KL deployment owner. No sponsor, funder or supporting organisation for the KL student implementation is stated. The KL deployment stakeholder and operational decision owner must be confirmed before approval.

FINAL ACHIEVEMENT REQUIRED

Deliver an interpretable cross-country poverty model with explicit targeting-error and interpretability trade-off analysis.

REFERENCE TO REPRODUCE IN CP1

Test cross-country generalisation where wealth distributions differ. This is the O2 reference against which every Interpretable Cross-Country Poverty Estimation from Survey Data contribution must be compared.

DIFFERENTIATING CONTRIBUTION IN CP2

Build interpretable models and quantify the accuracy cost of interpretability.

COMPLETE FINAL DEMONSTRATION

Analyse exclusion versus inclusion errors as distinct targeting costs. The final demonstration must show the complete Interpretable Cross-Country Poverty Estimation from Survey Data workflow rather than an isolated component.

INDEPENDENT ACCEPTANCE

Validate accuracy, generalisation and interpretability for policy use. The result must be independently checked under representative, boundary and failure conditions.

WHY THIS IS CAPSTONE-WORTHY

This is a two-semester culminating project because the team must establish the O2 reference “Test cross-country generalisation where wealth distributions differ”, design and evaluate the O3 contribution “Interpretable models and quantify the accuracy cost”, integrate the O4 system “Exclusion versus inclusion errors as distinct targeting costs”, and defend the O5 result under independent, boundary and failure conditions.

Problem evidence and approval condition**DIRECT CHALLENGE OR BENCHMARK BRIEF**

Evidence locator: Household poverty prediction · **Open cited source**

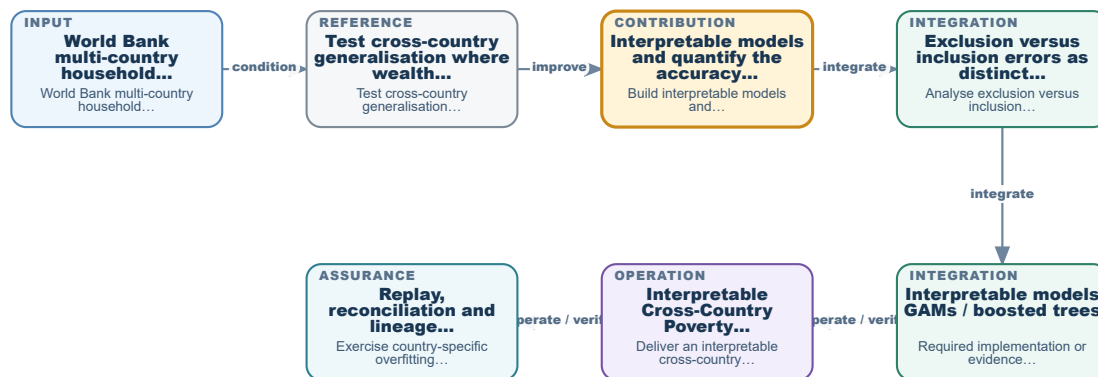
What the source establishes: The cited source establishes a reported problem, challenge, benchmark or technical requirement that can be reproduced and bounded for this project.

What the team must verify: Confirm the cited status and affected version, reproduce the reported behaviour or benchmark, and record any difference between the source and the approved project scope.

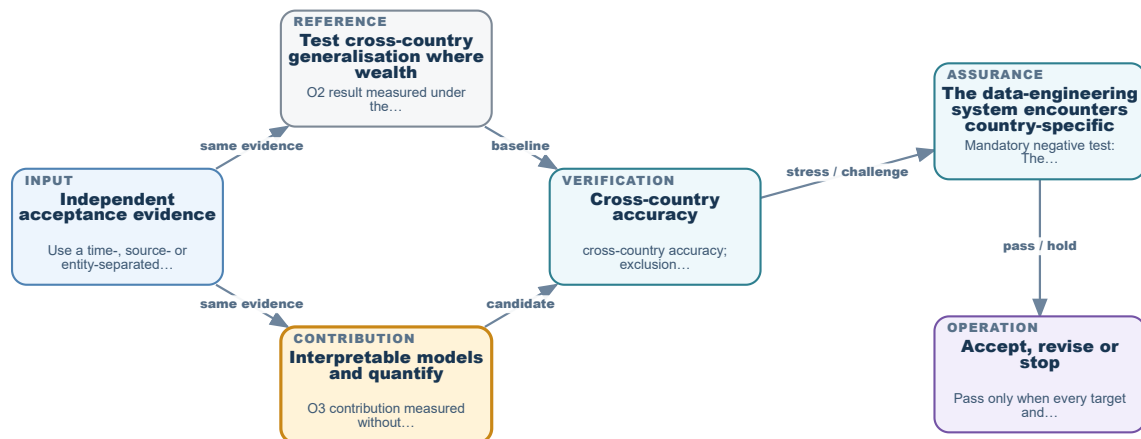
Pictorial project specification

data lineage

A. System / physical architecture — what the team must build and connect



B. Validation architecture — how the team must prove the result



Two-semester execution roadmap

Semester VII: problem + reference · Semester VIII: contribution + system + acceptance

SEMESTER VII / CAPSTONE PROJECT I

01

Validate the problem and freeze the engineering contract

Build poverty classification per country and measure baseline accuracy. Verify the beneficiary, source claim, system boundary, evidence access, interfaces, hazards and acceptance rules before implementation proceeds.

CP1 Gate 1 — the problem, scope, evidence and acceptance plan are approved.

Mandatory outcome and evidence

A per-country poverty baseline. Required evidence: source-to-claim map, stakeholder or reproduction evidence, requirements, architecture decision record, risk register and source-to-result lineage, schema and time semantics, quality rules and idempotent replay evidence.

02

Build the reproducible reference and vertical slice

Test cross-country generalisation where wealth distributions differ. Compare at least two feasible reference approaches, select one through a documented trade-off and measure it under AC-1 and AC-4.

CP1 Gate 2 — a reproducible reference and an integrated vertical slice are demonstrated.

Mandatory outcome and evidence

Cross-country generalisation results. Deliver executable fixtures, an independent oracle, unit tests, resource profile, failure log and baseline values for cross-country accuracy; exclusion vs inclusion error; interpretability cost; end-to-end lineage coverage (%).

SEMESTER VIII / CAPSTONE PROJECT II

03

Engineer and prove the differentiating contribution

Build interpretable models and quantify the accuracy cost of interpretability. Compare it with O2 using the same evidence and resource envelope; include ablation, sensitivity, trade-off or controlled experiments appropriate to the project.

CP2 Gate 3 — the contribution is technically defensible and its trade-offs are measured.

Mandatory outcome and evidence

A measured interpretability trade-off. Report baseline-versus-candidate results, uncertainty, condition-wise failures and a defensible positive or negative conclusion. Retain the selected configuration, experimental controls, raw evidence and a trace from each claim to its acceptance result.

04

Integrate the complete operable system

Analyse exclusion versus inclusion errors as distinct targeting costs. Connect the evidence path, reference and contribution, user or operator interface, configuration, security or safety controls, telemetry, degraded mode and recovery.

CP2 Gate 4 — the complete system operates normally, degrades safely and recovers.

Mandatory outcome and evidence

Asymmetric targeting-error analysis. Every component and interface must have unit, integration and system evidence in a representative environment. Retain configuration, interface contracts, telemetry, degraded-mode

behaviour and recovery proof.

05

Complete independent acceptance and impact validation

Validate accuracy, generalisation and interpretability for policy use. Execute AC-1 to AC-4 and all 5 negative tests; obtain independent domain, stakeholder or technical review.

CP2 Final Gate — every target and guardrail passes, or the project is explicitly held or revised.

Mandatory outcome and evidence

A policy-oriented validation report. Publish baseline, target and achieved values, limitations, residual risks, reproducibility package, operating guide, final report, demonstration video and individual contribution evidence.

Functional requirements

- FR-1 — Ingest or acquire “World Bank multi-country household survey data”; validate version, provenance, schema or protocol, units and project-specific quality gates before use.
- FR-2 — Reproduce “Test cross-country generalisation where wealth distributions differ” through a one-command, one-protocol or one-test-run reference path with an independent expected-result oracle.
- FR-3 — Implement “Interpretable models and quantify the accuracy cost” as a separately testable subsystem whose configuration and evidence are versioned.
- FR-4 — Implement and verify “Exclusion versus inclusion errors as distinct targeting costs” and its input, output, error, configuration and telemetry contracts.
- FR-5 — Implement and verify “Interpretable models GAMs / boosted trees” and its input, output, error, configuration and telemetry contracts.
- FR-6 — Deliver “Interpretable Cross-Country Poverty Estimation from Survey Data” as the complete demonstrable result; no isolated notebook, model, function or dashboard counts as completion.
- FR-7 — Detect the condition described by NT-1, invoke the documented safe or degraded response and retain recovery evidence.

Non-functional and domain-assurance requirements

- NFR-1 — Performance: meet every accepted KPI target under AC-1 and AC-4, including tail values and resource cost where applicable.
- NFR-2 — Reliability and recovery: pass every mandatory negative test; a critical failure cannot be hidden by an average.
- NFR-3 — Domain assurance: Every published result shall retain source-to-result lineage, schema and event-time semantics.
- NFR-4 — Evidence quality: Backfill, retry and replay shall be idempotent and reconciliation shall detect silent divergence.
- NFR-5 — Maintainability: use version control, automated tests, documented interfaces, configuration validation and a clean-environment build or protocol.
- NFR-6 — Operability: provide logs, measurements, health or quality indicators, operator guidance, degraded mode and a verified recovery procedure.

System boundary and permitted scope

INCLUDED

- Reproducible reference: Test cross-country generalisation where wealth distributions differ. This is the O2 reference against which every Interpretable Cross-Country Poverty Estimation from Survey Data contribution must be compared
- Differentiating contribution: Build interpretable models and quantify the accuracy cost of interpretability
- Integrated result: Analyse exclusion versus inclusion errors as distinct targeting costs. The final demonstration must show the complete Interpretable Cross-Country Poverty Estimation from Survey Data workflow rather than an isolated component
- Independent acceptance: Validate accuracy, generalisation and interpretability for policy use. The result must be independently checked under representative, boundary and failure conditions

EXPLICITLY EXCLUDED

- out of scope.
- A notebook, isolated model, single-function prototype, dashboard-only demonstration or unvalidated library integration.
- Production certification, universal generalisation and unattended high-stakes operation outside the validated envelope.
- An isolated notebook, model, function, dashboard or library integration without the complete operating and validation workflow.

DEPENDENCY ASSUMPTION

Access to every required dataset, sample, device, tool, reviewer, licence and approval must be confirmed at the project-approval review; unavailable dependencies require an approved scope revision.

Team, prerequisites, tools and feasibility

RECOMMENDED TEAM

A team of 3–4 students must assign named responsibility for source data and lineage leadership, pipeline or storage engineering leadership, serving and observability leadership, reconciliation and data quality leadership in Interpretable Cross-Country Poverty Estimation from Survey Data; each member must own reviewable implementation and evidence.

PREREQUISITES

Tabular ML, interpretability, survey data.

PRINCIPAL TOOLS

- interpretable models (GAMs / boosted trees with attribution)
- cross-country validation
- error-cost analysis
- versioned schema and fixture tools

FACILITY / COMPUTE

A reproducible local or cloud data platform with versioned fixtures, lineage, quality and replay tooling. Principal access must include interpretable models (GAMs / boosted trees with attribution), cross-country validation, error-cost analysis, versioned schema and fixture tools.

FALLBACK PATH

Use privacy-safe synthetic or public fixtures with the same schema, time, quality and scale characteristics required by the problem. It may reduce scale but must retain O2, O3, O4 and O5.

Acceptance conditions	
AC-1	Representative operation Validate accuracy, generalisation and interpretability for policy use.
AC-2	Boundary and failure operation Exercise country-specific overfitting; black-box models unusable for policy; treating both error types as equal.
AC-3	Independent acceptance evidence Use a time-, source- or entity-separated acceptance partition and reconcile every published result to source lineage.
AC-4	Frozen resource envelope Use the same approved versions, workload, hardware or facility limits and resource budget for reference and final comparisons; principal tools are interpretable models (GAMs / boosted trees with attribution), cross-country validation, error-cost analysis, versioned schema and fixture tools.

KPI baseline, target, method and pass contract

ID	MEASURE	BASELINE	DIRECTION	UNIT	TARGET / PASS RULE	CONDITIONS	MEASUREMENT METHOD	GUARDRAIL
KPI-1	cross-country accuracy	O2 reference measured with the same evidence, workload and resource envelope	higher is better	%	$\geq$ O2 reference + 3 percentage points, or within 1 point while improving the primary resource KPI by $\geq$ 20%	AC-1, AC-3	Use versioned fixtures and an independent oracle; report condition-wise values, repeated trials and uncertainty.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted
KPI-2	exclusion vs inclusion error	O2 reference measured with the same evidence, workload and resource envelope	lower is better	dimensionless score on a 0–100 rubric frozen before O2	$\geq$ 80/100 and at least 5 points higher than the O2 reference	AC-1, AC-3	Before O2, freeze a 0–100 rubric with at least five observable criteria and named 0, 50, 80 and 100 anchors; use two independent raters and report agreement plus condition-wise scores.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted
KPI-3	interpretability cost	O2 reference measured with the same evidence, workload and resource envelope	lower is better	₹ or approved currency	$\leq 0.80 \times$ O2 reference per accepted result, with no critical quality regression	AC-1, AC-4	Measure total and per accepted result, including compute, storage, network, material or energy overhead applicable to the system boundary.	The primary correctness, safety or quality metric may not regress by more than the approved non-inferiority margin.
KPI-4	end-to-end lineage coverage (%)	O2 reference measured with the same evidence, workload and resource envelope	higher is better	%	$\geq$ 95%; 100% for safety-, security- or compliance-critical items	AC-1, AC-3	Use versioned fixtures and an independent oracle; report condition-wise values, repeated trials and uncertainty.	No critical negative test, data-integrity failure or unapproved requirement regression is permitted
KPI-5	reconciliation error count	O2 reference measured with the same evidence,	lower is better	count	= 0 in every mandatory and adversarial condition	AC-2, AC-3	Execute every mandatory, boundary and adversarial fixture against an explicit	No critical negative test, data-integrity failure or unapproved

ID	MEASURE	BASELINE	DIRECTION	UNIT	TARGET / PASS RULE	CONDITIONS	MEASUREMENT METHOD	GUARDRAIL
		workload and resource envelope					expected-state oracle; retain complete logs and state evidence.	requirement regression is permitted
KPI-6	pipeline completion latency p95 (s)	O2 reference measured with the same evidence, workload and resource envelope	lower is better	s	$p_{95} \leq 0.80 \times O2$ reference at equal or better functional quality	AC-1, AC-4	Warm the system, run at least 30 measured trials per condition with a monotonic clock, and report p50, p95, p99 and failures.	The primary correctness safety or quality metric may not regress by more than the approved non-inferiority margin.

Mandatory negative-test and recovery campaign

NT- 1 The data-engineering system encounters country-specific overfitting during the Interpretable Cross-Country Poverty Estimation from Survey Data operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The data-engineering system encounters country-specific overfitting during the Interpretable Cross-Country Poverty Estimation from Survey Data operating or validation workflow”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT- 2 The data-engineering system encounters black-box models unusable for policy during the Interpretable Cross-Country Poverty Estimation from Survey Data operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The data-engineering system encounters black-box models unusable for policy during the Interpretable Cross-Country Poverty Estimation from Survey Data operating or validation workflow”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT- 3 The data-engineering system encounters treating both error types as equal during the Interpretable Cross-Country Poverty Estimation from Survey Data operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The data-engineering system encounters treating both error types as equal during the Interpretable Cross-Country Poverty Estimation from Survey Data operating or validation workflow”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT- 4 The data-engineering system encounters A backfill, retry or partial failure produces a non-idempotent result in Interpretable Cross-Country Poverty Estimation from Survey Data during the Interpretable Cross-Country Poverty Estimation from Survey Data operating or validation workflow.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “The data-engineering system encounters A backfill, retry or partial failure produces a non-idempotent result in Interpretable Cross-Country Poverty Estimation from Survey Data during the Interpretable Cross-Country Poverty Estimation from Survey Data operating or validation workflow”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

NT- 5 Data quality is measured globally while a critical partition remains defective in Interpretable Cross-Country Poverty Estimation from Survey Data.

Trigger: Create a controlled fixture, sample, workload or operating condition that reproduces this failure: “Data quality is measured globally while a critical partition remains defective in Interpretable Cross-Country Poverty Estimation from Survey Data”.

Expected safe behaviour: The invalid record or state transition must be detected before silent commit; replay must be idempotent and lineage must remain complete.

Recovery: Quarantine invalid data, repair from the versioned source or checkpoint, replay idempotently and reconcile every downstream result.

Evidence: Retain fixture or sample ID, precondition, version and configuration, expected versus observed result, logs or measurements, recovery time and residual-risk decision.

Evidence and reproducibility contract

PRIMARY EVIDENCE

World Bank multi-country household survey data with poverty labels

PROTOCOL AND QUALITY

Version the cited evidence and project inputs, reproduce “Test cross-country generalisation where wealth distributions differ”, implement “Build interpretable models and quantify the accuracy cost of interpretability”, and compare both under AC-1 to AC-4 without changing the approved evidence or resource envelope. Check schema, null/duplicate rules, event time, source-to-output lineage, replay consistency and reconciliation totals.

PROVENANCE AND INDEPENDENT PARTITION

The Interpretable Cross-Country Poverty Estimation from Survey Data evidence manifest must record source URL, access date, licence or permission, owner/custodian, record/sample/fixture identifiers, exclusions, transformations, configuration and the objective or KPI using each item. Reserve the AC-3 evidence before O3 tuning and separate it by the independent unit appropriate to data-platform engineering; no transformation may leak acceptance information into development.

REPRODUCIBILITY

A clean environment must rebuild the O2 reference, O3 contribution, integrated system and acceptance evidence for Interpretable Cross-Country Poverty Estimation from Survey Data using the version-pinned toolchain (interpretable models (GAMs / boosted trees with attribution), cross-country validation, error-cost analysis, versioned schema and fixture tools), seeds/configuration and an automated or fully scripted runbook.

Mandatory final deliverables

- D1 — Approved Interpretable Cross-Country Poverty Estimation from Survey Data problem charter, requirements, system boundary, architecture and source-to-claim map.
- D2 — Reproducible Test cross-country generalisation where wealth distributions differ with fixtures or protocol, oracle, tests, baseline results and resource profile.
- D3 — Independently testable Interpretable models and quantify the accuracy cost with design rationale, comparative evidence, trade-offs and limitations.
- D4 — Complete Interpretable Cross-Country Poverty Estimation from Survey Data prototype or experimental system with configuration, interfaces, telemetry, degraded operation and recovery.
- D5 — Automated or documented acceptance harness covering AC-1 to AC-4, all KPIs and every negative test.
- D6 — Versioned repository or protocol package, evidence manifest, installation/operating guide, API/interface documentation and reproducibility instructions.
- D7 — Final report, research paper or technical article, poster, portfolio case study, demonstration video, video logs and individual contribution/viva evidence.